

Phishing and Smishing Awareness: Prevention and Mitigation Strategies.

Author: Noel Job C. Silveron
Course: BS-Electrical Engineering
Date: August 20, 2026

I. Introduction

In the digital age, cyber threats have become increasingly sophisticated, targeting individuals through deceptive communication channels. Among the most prevalent and dangerous of these threats are phishing and smishing attacks, which exploit human trust to steal sensitive information such as passwords, credit card numbers, and personal identification data. Phishing attacks are delivered via email, while smishing attacks use SMS or text messages, but both share the same malicious intent: to deceive recipients into revealing confidential information or clicking on harmful links.

The impact of these attacks extends beyond financial loss, affecting individuals' privacy, mental well-being, and overall digital security. Students, working professionals, and even large organizations have fallen victim to these schemes, resulting in data breaches, identity theft, and reputational damage. As reliance on digital communication grows, the need for effective prevention strategies becomes increasingly critical. This report aims to define phishing and smishing, identify their key characteristics, and propose practical, systematic measures to prevent and mitigate their effects.

II. Problem Description

Phishing and smishing attacks have become a significant concern in today's interconnected world. Cybercriminals frequently impersonate trusted entities such as banks, government agencies, or known contacts to manipulate individuals into disclosing sensitive information. According to cybersecurity reports, millions of phishing attempts occur daily, with a substantial portion successfully breaching personal and organizational defenses.

The problem is particularly acute among students and working professionals who frequently use email and mobile devices for communication. Many individuals lack awareness of the warning signs, making them vulnerable to sophisticated scams. Furthermore, the increasing use of mobile devices for work and study has expanded the attack surface for smishing, as text messages often appear more personal and urgent than emails, leading to higher click-through rates.

Some of the key challenges identified include:

- **Lack of Awareness:** Many users cannot distinguish between legitimate and fraudulent messages.
- **Urgency and Manipulation:** Attackers create a false sense of urgency to pressure victims into acting quickly without verification.
- **Technical Sophistication:** Fake websites and messages increasingly mimic legitimate ones, making detection difficult.
- **Insufficient Security Practices:** Weak passwords and lack of multi-factor authentication (MFA) increase vulnerability.

With these challenges in mind, it is essential to propose structured, practical solutions that empower individuals to recognize, resist, and report phishing and smishing attempts effectively.

III. Proposed Solution

The proposed solution to mitigate phishing and smishing risks is a comprehensive awareness and prevention framework based on five key control measures. These measures are designed to be simple, actionable, and effective for students, employees, and general users.

Key Prevention Measures:

Measure Description

- **Block Spam Messages** Enable carrier spam protection and advanced email filtering to reduce exposure to malicious messages.
- **Double-Check Messages Through Trusted Channels** Verify unexpected alerts by contacting the organization through official channels (e.g., calling the number on the back of a bank card).
- **Never Click Links—Type Manually** Avoid clicking links in messages; instead, manually type the official URL into a browser or use the official app.
- **Change Passwords Regularly & Avoid Reuse** Update passwords periodically and never reuse the same password across multiple accounts.
- **Enable Multi-Factor Authentication (MFA)** Use app-based authenticators or hardware keys for an additional layer of security; avoid SMS-based MFA when possible.

These measures collectively create a robust defense against phishing and smishing attacks by addressing both technical and behavioral vulnerabilities. The framework emphasizes proactive habits, verification protocols, and the use of security tools to minimize risk.

IV. Conclusion

Phishing and smishing remain among the most common and effective cyber threats facing individuals and organizations today. Their success relies on human error and the exploitation of trust, making awareness and prevention critical. This report has outlined five practical measures—spam blocking, verification through trusted channels, manual URL entry, password hygiene, and multi-factor authentication—that collectively form a strong defense against these attacks.

By adopting these practices, students, working professionals, and the general public can significantly reduce their vulnerability to cyber fraud. Regular training, combined with the implementation of these controls, fosters a culture of security awareness that protects not only personal information but also organizational assets. Ultimately, proactive prevention is the most effective strategy against the growing threat of phishing and smishing.

V. References:

CISA. (n.d.). Avoid Phishing Scams. Cybersecurity & Infrastructure Security Agency.

<https://www.cisa.gov/avoid-phishing-scams>

Federal Trade Commission. (n.d.). How to Recognize and Avoid Phishing Scams.

<https://www.ftc.gov/phishing>

Kaspersky. (n.d.). What is Smishing? Definition and Prevention.

<https://www.kaspersky.com/resource-center/definitions/smishing>

WebMD. (n.d.). Burnout: Symptoms, Risk Factors, Prevention, Treatment.

<https://www.webmd.com/mental-health/burnout-symptoms-signs>